Doob's L2 Maximal Inequality

An End-to-End Meta-Math Walkthrough

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Abstract

Partition the first crossing of a threshold by its time. Conditional expectation transfers that crossing value to the terminal absolute value; integrating thresholds produces the L^2 estimate. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

PS-04 / Probability and Stochastic Processes / Report v1

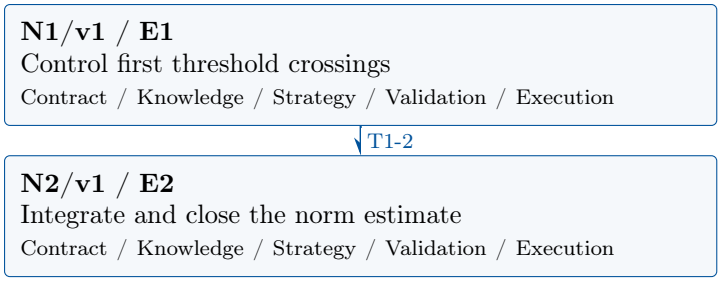
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *For a real square-integrable martingale $(M_k, \mathcal{F}_k)_{k=0}^N$ at a deterministic finite horizon, prove $\mathbb{E}[\max_{0\le k\le N} |M_k|^2] \le 4\mathbb{E}|M_N|^2$. The maximum includes the initial value, and the constant is independent of N.*

A. Foundations and permitted imports

Use the tower and pull-out properties of conditional expectation, $|\mathbb{E}[Z \mid \mathcal{F}]| \le \mathbb{E}[|Z| \mid \mathcal{F}]$, Cauchy–Schwarz, and Tonelli for nonnegative integrands. The tail formula $\mathbb{E}Z^2 = \int_0^\infty 2\lambda\mathbb{P}(Z > \lambda)d\lambda$ follows by integrating $Z^2 = \int_0^Z 2\lambda d\lambda$ and applying Tonelli. No optional-stopping theorem is needed for the proof.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. $\lambda \mathbb{P}(Y^* > \lambda) \leq \mathbb{E}[Y_N 1_{\{Y^* > \lambda\}}]$.

Receiving activity. N2.1

Compatibility. E1/v1, finite horizon, and square integrability of every M_k. Check: Y^* is in L^2 before division because $(Y^*)^2 \leq \sum_{k=0}^{N} |M_k|^2$; this avoids a circular finiteness argument.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Control first threshold crossings

Object and role. Use disjoint first-crossing events, each measurable at its crossing time. **Local obligation:** $\lambda\mathbb{P}(Y^* > \lambda) \le \mathbb{E}[Y_N 1_{\{Y^*>\lambda\}}]$.

Inputs. $Y_k = |M_k|$, $Y^* = \max_{0\le k\le N} Y_k$, and $\lambda > 0$.

C — Contract

Publish E1/v1 only after establishing this obligation: $\lambda\mathbb{P}(Y^* > \lambda) \le \mathbb{E}[Y_N 1_{\{Y^*>\lambda\}}]$.

K — Knowledge

$Y_k = |M_k|$, $Y^* = \max_{0\le k\le N} Y_k$, and $\lambda > 0$. Retain the stated assumptions and import versions.

S — Strategy

Use disjoint first-crossing events, each measurable at its crossing time.

V — Validation

For each k, $\mathbb{E}[Y_N \mid \mathcal{F}_k] \ge Y_k$ follows from the martingale property and conditional absolute-value inequality.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 Let $A_k = \{Y_0 \le \lambda, \dots, Y_{k-1} \le \lambda, Y_k > \lambda\}$, with the previous conditions empty for $k = 0$. These disjoint events lie in $\mathcal{F}_k$ and their union is $\{Y^* > \lambda\}$. On A_k, $Y_k > \lambda$.

N1.2 By conditional expectation, $\mathbb{E}[Y_N 1_{A_k}] = \mathbb{E}[\mathbb{E}(Y_N \mid \mathcal{F}_k)1_{A_k}] \ge \mathbb{E}[Y_k 1_{A_k}] \ge \lambda\mathbb{P}(A_k)$. Summing the finitely many inequalities gives the export.

Export E1/v1. $\lambda\mathbb{P}(Y^* > \lambda) \le \mathbb{E}[Y_N 1_{\{Y^*>\lambda\}}]$.

Acceptance gate. For each k, $\mathbb{E}[Y_N \mid \mathcal{F}_k] \ge Y_k$ follows from the martingale property and conditional absolute-value inequality.

Targeted repair. If the whole crossing event is treated as measurable at every time, partition it by first-crossing time before applying conditional expectation. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Integrate and close the norm estimate

Object and role. Integrate the tail bound and apply Cauchy–Schwarz, justifying division by the maximal norm. **Local obligation:** $\|Y^*\|_2 \le 2\|M_N\|_2$, hence the claimed squared bound.

Inputs. E1/v1, finite horizon, and square integrability of every M_k. Incoming proof exports: E1/v1 at N2.1.

C — Contract

Publish E2/v1 only after establishing this obligation: $\|Y^*\|_2 \le 2\|M_N\|_2$, hence the claimed squared bound.

K — Knowledge

E1/v1, finite horizon, and square integrability of every M_k. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

S — Strategy

Integrate the tail bound and apply Cauchy–Schwarz, justifying division by the maximal norm.

V — Validation

Y^* is in L^2 before division because $(Y^*)^2 \le \sum_{k=0}^{N} |M_k|^2$; this avoids a circular finiteness argument.

E — Execution

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 The finite-sum bound makes $A = \|Y^*\|_2$ finite. Multiply E1 by 2 and integrate λ over $(0, \infty)$. The tail formula and Tonelli give $A^2 \le 2\int_0^\infty \mathbb{E}[Y_N 1_{\{Y^* > \lambda\}}]d\lambda = 2\mathbb{E}[Y_N Y^*]$.

N2.2 Cauchy–Schwarz bounds the final expression by $2\|Y_N\|_2 A$. If $A = 0$, the conclusion is immediate. Otherwise divide by A to obtain $A \le 2\|Y_N\|_2$ and square. The factor four and horizon independence follow.

Export E2/v1. $\|Y^*\|_2 \le 2\|M_N\|_2$, hence the claimed squared bound.

Acceptance gate. Y^* is in L^2 before division because $(Y^*)^2 \le \sum_{k=0}^{N} |M_k|^2$; this avoids a circular finiteness argument.

Targeted repair. If division is performed before maximal integrability is known, first use the finite sum of squared values, then handle the zero-norm case separately. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Control first threshold crossings

1. **N1.1.** Let $A_k = \{Y_0 \le \lambda, \dots, Y_{k-1} \le \lambda, Y_k > \lambda\}$, with the previous conditions empty for $k = 0$. These disjoint events lie in $\mathcal{F}_k$ and their union is $\{Y^* > \lambda\}$. On A_k, $Y_k > \lambda$.
2. **N1.2.** By conditional expectation, $\mathbb{E}[Y_N 1_{A_k}] = \mathbb{E}[\mathbb{E}(Y_N \mid \mathcal{F}_k)1_{A_k}] \ge \mathbb{E}[Y_k 1_{A_k}] \ge \lambda \mathbb{P}(A_k)$. Summing the finitely many inequalities gives the export.

B. N2 — Integrate and close the norm estimate

1. **N2.1.** The finite-sum bound makes $A = \|Y^*\|_2$ finite. Multiply E1 by 2 and integrate λ over $(0, \infty)$. The tail formula and Tonelli give $A^2 \le 2 \int_0^\infty \mathbb{E}[Y_N 1_{\{Y^* > \lambda\}}] d\lambda = 2\mathbb{E}[Y_N Y^*]$.
2. **N2.2.** Cauchy–Schwarz bounds the final expression by $2\|Y_N\|_2 A$. If $A = 0$, the conclusion is immediate. Otherwise divide by A to obtain $A \le 2\|Y_N\|_2$ and square. The factor four and horizon independence follow.

□

V. WORKED ORCHESTRATION

For a symmetric random walk with two increments and $M_0 = 0$, paths $++$ and $--$ have squared maximum four, while $+-$ and $-+$ have squared maximum one. Thus $\mathbb{E}(Y^*)^2 = (4 + 4 + 1 + 1)/4 = 5/2$. The terminal second moment is two, and $5/2 \le 4 \cdot 2 = 8$. The bound controls the whole path, not merely its endpoint.

A. Boundary and hypothesis audit

Adaptation alone is insufficient. The deterministic process $M_0 = 1, M_1 = 0$ is adapted and square-integrable but not a martingale. Its maximal square is one while its terminal square is zero, contradicting the claimed inequality without the martingale hypothesis.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected N1 inequality: $\mathbb{E}[Y_N \mid \mathcal{F}_k] \ge Y_k$ holds for any adapted process. The deterministic decay example disproves it. **Repair:** derive it from $M_k = \mathbb{E}[M_N \mid \mathcal{F}_k]$ and the

conditional absolute-value inequality, exactly as the import contract allows for a martingale. Recheck the crossing bound and integrated estimate under those hypotheses.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Use the inequality to control convergence of martingale approximations or construct stochastic integrals in L^2.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. Amir Dembo, Probability Theory: STAT310/MATH230 notes for measure-theoretic probability and limit arguments; Stanford STAT310C syllabus and lecture materials for martingale inequalities and stopping theorems. The versions and proofs used in this collection are stated explicitly.